

Mani Malek Esmaeili

CONTACT INFORMATION

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Code: <https://github.com/Anonymani?tab=stars>

SUMMARY

Tech Lead and Research Scientist at Google DeepMind focusing on AGI readiness, with research interests at the intersection of AI safety, security, and interpretability. Concurrently exploring black-box and mechanistic interpretability to bridge the gap between observable model behaviors and their underlying neural mechanisms. Currently studying continual learning systems, with a particular focus on emerging misbehaviors.

KEYWORDS AND HIGHLIGHTS

- **AI Safety & Alignment, Interpretability (Black-box & Mechanistic), Automated Red-Teaming, Post-Training (SFT, RL), Model Steering, Agentic Systems, Continual Learning, Private Machine Learning, Differential Privacy, Federated Learning, NLP, Computer Vision, Recommendation Systems, Information Retrieval, Signal Processing, Image/Video Processing, Audio Processing, Data Processing.**
- Research Scientist with a solid record of delivering AI to production.
- Highly experienced in initiating and leading projects across multiple teams.
- Track record of mentoring junior and senior engineers in Industry and Academia.

INDUSTRY EXPERIENCE

Google DeepMind

Tech Lead and Research Scientist

Working on automated red-teaming, post-training (SFT, RL), black-box interpretability, mechanistic interpretability, model steering, agentic systems, and continual learning.

Nov. '23 - present

Meta (Facebook)

Technical Lead Manager, Reality Labs, Meta

Leading private ML initiatives in AR/VR.

Setting future directions for private insights in metaverse.

Managing a small and elite team of scientists and engineers.

Jan. '23 - Nov. '23

Lead Research Scientist in privacy in ML, Meta AI

Initiated and led **Federated Learning** research in applied AI org, which resulted in multiple influential product launches.

Lead multiple **open sourcing** efforts within ML privacy.

Proposed and led various research efforts in **private ML**.

Feb. '20 - Dec. '22

Software Engineer and Tech lead, Video Infra

Developed highly scalable **Audio/Video retrieval** algorithms.

Lead the research in video/audio understanding algorithms.

Apr. '17 - . Jan. '20

Magnitude Software (Simba Technologies), Vancouver, Canada

Developed SQL language parsers and implemented various SQL APIs.

Developed efficient algorithms for **data processing at scale**.

Dec. '15 - . Mar. 17

Sasset Technology Inc., Vancouver, Canada

Founded the company to improve EQ safety using computer vision.

Designed and developed the full software stack.

Managed all engineers.

Jul. '13 - . Nov. 15

Rigid Robotics Inc., Vancouver, Canada (Contract-student)

Developed their **object tracking and segmentation** module.

Jul. '13 - Feb. '14

Sidebuy Tech. Inc., Vancouver, Canada (Contract-student)

Developed their state-of-the-art **image search** module.
Developed an **integrated text/image retrieval** platform.

Jul. '12 - Feb. '13

BroadbandTV Corp., Vancouver, Canada (Contract-student)

Developed an integrated **audio/video copy retrieval system**.

Jul. '10 - Mar. '11

EDUCATION

The University of British Columbia, Vancouver, BC

- PhD, Electrical and Computer Engineering (GPA: 90)
- Sub-specialization, Engineering Management Program (GPA: 91)

Sep. '07 - May '13

Sep. '10 - Apr. '11

The University of Tehran, Tehran, Iran

- MSc, Electrical and Computer Engineering (GPA: 18 out of 20)
- BSc, Electrical and Computer Engineering (GPA: 17.5 out of 20)

Sep. '04 - Apr. '07

Sep. '00 - Sept. '04

ACADEMIC
EXPERIENCE

Professional Service & Activities

- Technical reviewer at NeurIPS. Jun. '20 - present
- Panelist on distributed computing and federated learning in IEEE CCNC Jan. '22.
- Representative of Facebook Industry Workshop in ICIP. Sep. '19
- Technical reviewer of various IEEE, Elsevier journals and conferences. Nov. '11 - Nov. '16
- Technical reviewer for IEEE T-IFS, IEEE J-STSP, Elsevier JVCI, Hindawi JAM, IEEE MMSP, and IEEE ISCAS. Nov. '11 - present
- Volunteer organizer of IEEE ICASSP 2013. May, '13
- Councilor at UBC Graduate Student Association. Mar. '08 - Oct. '08
- Member of the organizing team for UBC's Graduate Students Orientation. Aug. '07 - Sep. '07

Supervision, UBC, Vancouver, Canada

Co-supervised four undergraduate students at UBC through their final projects.

Jan. '10 - Aug. '11

Mentored three undergraduate students who joined our research lab at UBC as interns.

Teaching Assistant, UBC, Vancouver, Canada

Served as teaching assistant for the following courses:

DSP systems, Introduction to Computation, **Signals & Systems**, Engineering Electromagnetic, and Electronic circuits.

Sep. '08 - May. '12

Teaching Assistant, University of Tehran, Iran

Lab instructor and lecturer for **Electronic Circuits**.

Jun. '03 - Sep. '04

PUBLICATIONS

- Visual prompt engineering for video models. *arXiv preprint arXiv:2607.25537, 2026.*
- The ACUTE Protocol: Operationalizing Language Model Activations for Better Calibration, Utility, and Trust. *International Conference on Machine Learning (ICML), 2026.*
- Hiding in Plain Text: Detecting Concealed Jailbreaks via Activation Disentanglement. *arXiv preprint arXiv:2602.19396, 2026.*
- Interpreting and Controlling Model Behavior via Constitutions for Atomic Concept Edits. *International Conference on Artificial Intelligence and Statistics (AISTATS), 2026.*
- Concept Spaces in the Residual Stream of Diffusion Transformers. *CVPR Workshops, 2026.*
- ShieldGemma 2: Robust and Tractable Image Content Moderation. *arXiv preprint arXiv:2504.01081, 2025.*
- Joint Federated Learning and Personalization for on-Device ASR. *IEEE Automatic Speech Recognition and Understanding Workshop (ASRU), 2023.*
- Federated Ensemble Learning: Increasing the Capacity of Label Private Recommendation Systems. *IEEE Data Engineering Bulletin, 2023.*
- FEL: High Capacity Learning for Recommendation and Ranking via Federated Ensemble Learning. *arXiv preprint arXiv:2206.03852, 2022.*
- Federated Learning with Buffered Asynchronous Aggregation. *International Conference on Artificial Intelligence and Statistics (AISTATS), 2022.*
- PAPAYA: Practical, Private, and Scalable Federated Learning. *Conference on Machine Learning and Systems (MLSys), 2022.*
- Antipodes of Label Differential Privacy: PATE and ALIBI. *Advances in Neural Information Processing Systems (NeurIPS), 2021.*
- Opacus: User-Friendly Differential Privacy Library in PyTorch. *arXiv preprint arXiv:2109.12298, 2021.*
- A local fingerprinting approach for audio copy detection. *Signal Processing, 2014.*
- Multimedia copy detection. *Ph.D. Thesis, University of British Columbia, 2013.*
- Image and Video Copy Detection Using Content-Based Fingerprinting. *Multimedia Image and Video Processing (Book Chapter), 2012.*
- A novel local audio fingerprinting algorithm. *IEEE International Workshop on Multimedia Signal Processing (MMSP), 2012.*

- A Fast Approximate Nearest Neighbor Search Algorithm in the Hamming Space. *IEEE Transactions on Pattern Analysis and Machine Intelligence (T-PAMI)*, 2012.
- Fast matching for video/audio fingerprinting algorithms. *IEEE International Workshop on Information Forensics and Security (WIFS)*, 2011.
- A Robust and Fast Video Copy Detection System Using Content-Based Fingerprinting. *IEEE Transactions on Information Forensics and Security*, 2011.
- Fine localization of photons on the pixellated detectors based on the analytical model. *Canadian Conference on Electrical and Computer Engineering (CCECE)*, 2010.
- Robust video hashing based on temporally informative representative images. *International Conference on Consumer Electronics (ICCE)*, 2010.
- Video Copy Detection Using Temporally Informative Representative Images. *IEEE International Conference on Machine Learning and Applications (ICMLA)*, 2009.
- Topology Optimization for Backbone Wireless Mesh Networks. *Conference on Communication Networks and Services Research (CNSR)*, 2007.

SELECTED HONORS
AND AWARDS

- IEEE Hector J. MacLeod Scholarship Award for student achievement. Mar. '13
- The University of British Columbia Ph.D. tuition fee award. Sep. '07
- University of Tehran, faculty of Engineering Award of excellence. Aug. '04
- Ranked in top 0.1% in different Iranian nationwide university exams.